## Section 3: Special Distributions

Although we will introduce some special distributions in this chapter, we will start off with some familiar distributions starting with the binomial:

<u>Binomial Distribution:</u>
Let $X$ be a random variable that follows a binomial distribution, where $X \sim bin(n, p)$, then the PMF is defined as:

$$p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{e/w} \end{cases}$$

$$\binom{n}{x} \downarrow \frac{n!}{x!(n-x)!}$$

where the MGF is defined as:

$$M(t) = E(e^{tX}) = \sum e^{tX} \cdot \binom{n}{x} p^x (1-p)^{n-x}$$
$$= \sum \binom{n}{x} (p \cdot e^t)^x (1-p)^{n-x}$$

Taking advantage of the binomial theorem:

$$(a+b)^n = \sum \binom{n}{x} b^x \cdot a^{n-x}$$

We can see that the MGF can be defined as:

$$M(t) = \left[(1-p) + pe^t\right]^n$$

With the MGF, we can now define the first and second moments to determine the mean and variance of random variable that follows a binomial distribution.

$$M'(t) = \frac{d}{dt}\left[(1-p) + pe^t\right]^n$$
$$= n\left[(1-p) + pe^t\right]^{n-1} \cdot \frac{d}{dt}\left[(1-p) + pe^t\right]$$
$$= n\left[(1-p) + pe^t\right]^{n-1} \cdot pe^t$$

$$M'(t=0) = np = E(X)$$

$$M''(t) = \frac{d}{dt}\left[n\left[(1-p) + pe^t\right]^{n-1} \cdot pe^t\right]$$
$$= n(n-1)\left[(1-p) + pe^t\right]^{n-2} \cdot (pe^t)^2 + n\left[(1-p) + pe^t\right]^{n-1} \cdot pe^t$$

$$M''(t=0) = n(n-1)(p^2) + np = E(X)$$

$$Var(X) = E(X^2) - E(X)^2$$
$$= \left[n(n-1)(p^2) + np\right] - (np)^2$$
$$= np(1-p)$$

Example 3.1:

$$M(t) = \left(\frac{2}{3} + \frac{1}{3}e^{t}\right)^{5}$$

$$\Rightarrow X \sim bin\left(n=5, p=\frac{1}{3}\right)$$

$$E(x) = np = (5)\left(\frac{1}{3}\right) = \frac{5}{3}$$

$$Var(x) = np(1-p) = (5)\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = \frac{10}{9}$$

Example 3.2:

$$X \sim bin\left(n=n, p=\frac{1}{3}\right)$$

$$\begin{aligned} P(x \geq 1) &= \sum_{x=1}^{n} P(x) \\ &= 1 - P(x=0) \\ &= 1 - p(0) \\ &= 1 - \left[\binom{n}{0}\left(\frac{1}{3}\right)^{0}\left(\frac{2}{3}\right)^{n-0}\right] \\ &= 1 - \left(\frac{2}{3}\right)^{n} \end{aligned}$$

Find the minimum value of $n$ s.t. $P(x \geq 1) > 0.80$

$$1 - \left(\frac{2}{3}\right)^{n} > 0.8$$

$$\left(\frac{2}{3}\right)^{n} < 0.2$$

$$n \ln\left(\frac{2}{3}\right) < \ln(0.2)$$

$$n > \frac{\ln(0.2)}{\ln(2/3)}$$

$$n > 3.969$$

$$n \geq 4 \quad (n \text{ is discrete})$$

$$\boxed{n=4}$$

Negative Binomial Distribution ☆ NOT TESTABLE

Consider a sequence of independent Bernoulli experiments (similar to a Binomial distribution) but let $Y$ be the number of failures before the $r^{th}$ success (i.e. $Y + r = n$ trials). Then $Y \sim nbin(r, p)$, then the PMF is defined as:

$$p(y) = \begin{cases} \binom{y + [r-1]}{r-1} p^r \, (1-p)^y, & y = 0, 1, 2, \ldots \\ 0, & \text{e/w} \end{cases}$$

Notice that, unlike the "regular" binomial distribution, there is no finite limit for $y$.

Question: Why should this not be surprising?

We can see that the MGF can be defined as:

$$M(t) =$$

$\binom{-n}{k}$

Geometric Distribution: Special Case ☆ NOT TESTABLE

Let $r = 1$, then the negative binomial $Y \sim nbin(1, p)$ is a special case also known as the $Y \sim geom(p)$, where the support of $Y \in \mathbb{Z}^+$, which should not be surprising. Recall that a negative binomial involves the number of failures before the $r^{th}$ success, so the geometric is the number of failures before the 1st success, hence if we are successful on the 1st trial, we have no failures, ergo $Y = 0$.

The MGF is:

$$M(t) = \frac{p}{1 - (1 - p)e^t}$$

It is worth noting that the geometric distribution is one of two discrete probability distributions:

1. The probability distribution of the number $Y = X - 1$ of **failures** before the first success
2. The probability distribution of the number $X$ of **Bernoulli trials** needed to get one success, supported on the set $\mathbb{N}^+: \{1, 2, \ldots\}$.

The supports of $X$ and $Y$ are:

and the PMFs are:

Given the differing PMFs, then:

Real World Statistics: Geometric distributions

Methicillin-resistant Staphylococcus aureus (MRSA) is one of the leading causes of hospital-based infections and many models indicate that the distribution of the number of patients colonized at the time of detection of the index case is geometrically distributed. The probability of a MRSA case in a hospital is low, but not small, and tests for infection are routinely checked for antibiotic resistance. If the probability of a randomly selected sample is MRSA positive is $P(S) = 0.3$ and $Y$= the number of samples before the next MRSA positive,

and the expected number of samples before next positive is:

$$E(Y) = \frac{1-p}{p} = \frac{0.7}{0.3} = 2.\bar{3}$$

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Theorem 3.1.1: Let $X_1, X_2, \ldots, X_m \overset{iid}{\sim} bin(n_i, p)$ and let $Y = \sum X_i$ then $Y \sim bin(\sum_{i=1}^{m} n_i, p)$

independent and identically distributed

Proof: Using the MGF and Theorem 2.5.5 come in handy, as we know the MGF for $Y$ can be defined as:

$$M_Y(t) = E(e^{tY}) = E\left(e^{t\sum x_i}\right) = E\left(e^{tx_1 + tx_2 + \ldots + tx_m}\right)$$

$$= E\left(e^{tx_1} e^{tx_2} \cdots e^{tx_m}\right)$$

$$= \sum_{x_1} \sum_{x_2} \cdots \sum_{x_m} e^{tx_1} e^{tx_2} \cdots e^{tx_m} \cdot p(x_1, x_2, \cdots, x_m)$$

$$= \sum_{x_1} \sum_{x_2} \cdots \sum_{x_m} e^{tx_1} e^{tx_2} \cdots e^{tx_m} \cdot p(x_1) p(x_2) \cdots p(x_m)$$

☆ try for yourself

Note: Theorem 2.6.1 is more robust for the proof above; however, it is an extension of Theorem 2.5.5, which looks at $X_1, X_2$ being independent random variables, to several independent random variables.

Example 3.3:

Let $X \sim p(x)$ s.t. $M(t) = \left(\frac{1}{3} + \frac{2}{3}e^t\right)^5$

$\Longrightarrow X \sim bin\left(5, \frac{2}{3}\right)$

$$P(1.5 < X < 3.6) = P(2 \le X \le 3)$$
$$= P(X=2 \cup X=3)$$
$$= \sum_{x=2}^{3} \binom{5}{x}\left(\frac{2}{3}\right)^x\left(\frac{1}{3}\right)^{5-x}$$
$$= \frac{120}{243}$$

Example 3.4: Let $X \sim PMF \ni p(x) = \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^x$, where $x = 0, 1, 2, \ldots$, find the conditional PMF of $X$ given $X \geq 3$.

↳ This is $X \sim geom\left(p = \frac{1}{3}\right)$ but the steps/process are the focus

Let's break it into 2 parts:

(1) $P(X \geq 3) = 1 - P(X < 3)$
$$= 1 - P(X \le 2)$$
$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$
$$= \frac{8}{27} = \left(\frac{2}{3}\right)^3$$

(2) $P(X \cap X \geq 3)$ where the intersection produces a set or a null set

$$P(X \mid X \geq 3) = \frac{P(X \cap X \geq 3)}{P(X \geq 3)}$$
$$= \frac{\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^x}{(2/3)^3} \quad \text{where} \quad x = 3, 4, \ldots$$
$$= \begin{cases} \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^{x-3}, & x = 3, 4, 5, \ldots \\ 0, & e/w \end{cases}$$

Poisson Distribution

Let $X$ be a random variable that follows a Poisson distribution, where $X \sim Pois(\lambda)$, then the PMF is defined as:

$$p(x) = \begin{cases} \dfrac{\lambda^x \cdot e^{-\lambda}}{x!}, & x = 0, 1, 2, \ldots \\ 0, & \text{e/w} \end{cases}$$

where the MGF is defined as:

$$M(t) = E(e^{tX}) = \sum e^{tX} \cdot \frac{\lambda^x \cdot e^{-\lambda}}{x!} = \sum e^{tx} e^{-\lambda} \cdot \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} e^{tx} \cdot \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \sum^{\infty} \frac{(\lambda e^t)^x}{x!}$$

$$= e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

Recall the power series:

$$e^y = \sum_{x=0}^{\infty} \frac{y^x}{x!} = 1 + y + \frac{y^2}{2} + \frac{y^3}{6} + \cdots$$

With the MGF, we can now define the first and second moments to determine the mean and variance of random variable that follows a Poisson distribution.

$$M'(t=0) = \frac{d}{dt}\left[e^{\lambda(e^t - 1)}\right]\Big|_{t=0}$$

$$= \left[e^{\lambda(e^t - 1)} \cdot \lambda e^t\right]\Big|_{t=0}$$

$$= \lambda$$

$$M''(t=0) = \lambda^2 + \lambda$$

$$\Rightarrow \sigma^2 = Var(x) = E(x^2) - E(x)^2$$

$$= \lambda^2 + \lambda - \lambda^2$$

$$= \lambda$$

Example 3.5:

if $X \sim p(x)$ s.t. $M(t) = e^{4(e^t - 1)}$

$\Longrightarrow X \sim Pois(\lambda = 4)$

Theorem 3.2.1: Suppose $X_1, X_2, \dots, X_n \overset{iid}{\sim} Pois(\lambda_i)$. If $Y = \sum X_i$ then $Y \sim Pois(\lambda_Y = \sum \lambda_i)$

Proof: $Y = \sum X_i$, where $X_i \sim Pois(\lambda_i)$

$$M_Y(t) = E(e^{tY}) = E(e^{t\sum X_i}) = E(e^{tx_1} \cdot e^{tx_2} \cdot \dots \cdot e^{tx_n})$$

$$= E(e^{tx_1}) \cdot E(e^{tx_2}) \cdot \dots \cdot E(e^{tx_n})$$

$$= \prod_{i=1}^{n} E(e^{tx_i})$$

$$= \prod_{i=1}^{n} e^{\lambda_i (e^t - 1)}$$

$$= e^{\sum \lambda_i [e^t - 1]} \Rightarrow Y \sim Pois(\lambda_y = \sum \lambda_i)$$

Example 3.6: Let $X \sim Pois(\lambda) \ni P(X = 1) = P(X = 3)$, find the mode of $X$.

$$P(X=1) = \frac{\lambda^1 e^{-\lambda}}{1!} \qquad P(X=3) = \frac{\lambda^3 e^{-\lambda}}{3!}$$

$$P(X=1) = P(X=3)$$

$$\frac{\lambda e^{-\lambda}}{1!} = \frac{\lambda^3 e^{-\lambda}}{3!}$$

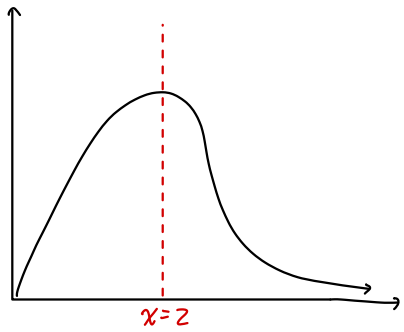
$$6\lambda = \lambda^3$$

$\lambda = \pm\sqrt{6}$ but $\lambda > 0$ by defn

$$\lambda = \sqrt{6}$$

mode = most frequently observed value among $x = 0, 1, 2, \dots$

| $x$ | $P(X=x)$ |
|---|---|
| 0 | 0.086 |
| 1 | 0.211 |
| 2 | 0.259 |
| 3 | 0.211 |
| 4 | 0.130 |
| $\vdots$ | $\vdots$ |
| $x\uparrow$ | $P(x)\downarrow$ |




$\text{mode} = 2 = \lfloor \lambda \rfloor$, $\qquad \lambda \in \mathbb{Z} \rightarrow \text{mode} = \lambda,\ \lambda - 1$